A Comparative Analysis of Classical Machine Learning Methods and Variational Quantum Methods for Breast Tumor Classification

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Outline

1. Classical Machine Learning Methods Introduction
2. Quantum Computing and Quantum Machine Learning Basics
   - The Qubit

Introduction

Our goals:

- Tumor Classification using Classical ML methods
- Tumor Classification using Variational Quantum methods
- Comparison of the results

Thesis outline

Workflow of the master thesis:

- Classical Machine Learning Methods Introduction
- Quantum Computing and Quantum Machine Learning Basics
- EDA & Classification using Classical ML methods and rnCV
- Classification using PCA and VQCs
- Comparison of the Classical and Quantum approaches

Classical Machine Learning Methods Introduction

Classical ML methods that we will be using:

- Logistic Regression
- Gaussian Naive Bayes
- Linear Discriminant Analysis
- Support Vector Machines
- Random Forests
- LightGBM

Quantum Computing and Quantum Machine Learning Basics

Key points:

- The Qubit
- Qubit Manipulation and Quantum Entanglement
- Quantum Gates
- Quantum Circuits
- Data Encoding
- Variational Quantum Classifiers

Bloch sphere

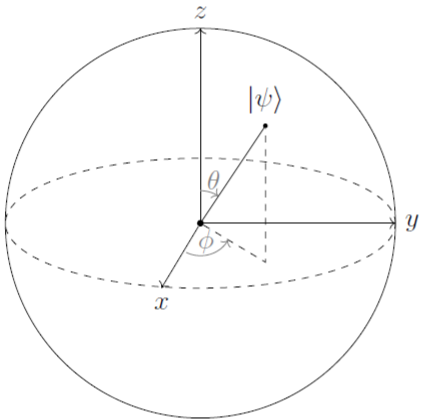

Figure: Bloch sphere representation of a qubit

Pauli Operators

$$\hat{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{1}$$

$$\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{2}$$

$$\hat{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{3}$$

$$\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{4}$$

Conclusion

VQCs show strong potential, especially in low dimensionality problems

Classical ML still slightly superior

Future quantum hardware will improve results

Thank you for your attention!

Questions?